

# Zachary Porat

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🖱 <https://zporat.github.io>  
🌐 U.S. Citizen  
📅 Last Updated September 4, 2026

## EDUCATION

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**Wesleyan University** Middletown, CT  
*Ph.D. in Mathematics* 2020 – 2026

- Thesis: “On the Cuspidal Cohomology of Congruence Subgroups of  $SL(3, \mathbb{Z})$ ” [[Link](#)]
- Advisor: David Pollack

**Union College** Schenectady, NY  
*B.S. in Mathematics with Honors, summa cum laude* 2016 – 2020

- Honors Thesis: “Classification of Torsion Subgroups for Mordell Curves” [[Link](#)]
- Physics Minor, Phi Beta Kappa, Scholars Program

## EMPLOYMENT

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**Bucknell University** Lewisburg, PA  
*Visiting Assistant Professor* 2026 – Present

## RESEARCH INTERESTS

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**Broad:** number theory, arithmetic geometry

**Specific:** automorphic forms, cohomology of arithmetic groups, arithmetic statistics, algebraic curves, computational aspects of number theory and arithmetic geometry

## PUBLICATIONS AND PREPRINTS

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- [6] **On the Cuspidal Cohomology of Iwahori Congruence Subgroups of  $SL(3, \mathbb{Z})$** , submitted (2026). [[Code](#)]

🔗 [arXiv:2609.01776](https://arxiv.org/abs/2609.01776)

- [5] **Distinguishing Elliptic Curves Modulo  $p$  and Identifying Images of Product Representations** (with Jessica Bennett, Jeffrey Hatley, Sasha Kononova, Vincent Macri, and Naina Praveen), submitted (2026). [[Code](#)]

🔗 [arXiv:2608.10880](https://arxiv.org/abs/2608.10880)

- [4] **Heuristics for (Ir)reducibility of p-Rank Strata of the Moduli Space of Hyperelliptic Curves** (with Thomas Bouchet, Erik Davis, Steven R. Groen, and Benjamin York), *Experimental Mathematics* (2026). [Code]  
 DOI: [10.1080/10586458.2026.2671126](https://doi.org/10.1080/10586458.2026.2671126)
- [3] **Computations Directly on the Cuspidal Cohomology of Congruence Subgroups of  $SL(3, \mathbb{Z})$** , *Mathematics of Computation* (2025). [Code]  
 DOI: [10.1090/mcom/4155](https://doi.org/10.1090/mcom/4155)
- [2] **Family Sizes for Complete Multipartite Graphs** (with Danielle Gregg, Thomas Mattman, and George Todd), *Involve, a Journal of Mathematics*, Vol. 15, No. 4 (2022), 669-686.  
 [MR4536581](https://www.ams.org/mathscinet/MR4536581), DOI: [10.2140/involve.2022.15.669](https://doi.org/10.2140/involve.2022.15.669)
- [1] **PIXE Analysis of Synthetic Turf** (with Sajju Chalise, Morgan Clark, Skye Conlan, Scott LaBrake, and Michael Vineyard), *Environment and Ecology Research*, Vol. 6, No. 1 (2018), 60-65.  
 DOI: [10.13189/eer.2018.060105](https://doi.org/10.13189/eer.2018.060105)

## TEACHING EXPERIENCE

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### Bucknell University

*Instructor of Record*

Fall 2026 – Present

- MATH 245 - Linear Algebra (Fall 2026)
- MATH 201 - Calculus I (Fall 2026  $\times 2$ )

### Wesleyan University

*Instructor of Record*

Fall 2022 – Spring 2026

- MATH 500 - Graduate Pedagogy (Fall 2025)
- MATH 121 - Calculus I (Fall 2024)
- MATH 132 - Elementary Statistics (Fall 2022)

*Graduate Teaching Assistant*

Fall 2021 – Spring 2026

- MATH 221 - Vectors and Matrices (Spring 2026)
- MATH 261 - Introduction to Abstract Algebra (Fall 2023, Fall 2025)
- MATH 225 - Introduction to Real Analysis (Spring 2022, Spring 2023, Spring 2025)
- MATH 262 - Advanced Topics in Abstract Algebra (Spring 2024)
- MATH 228 - Discrete Mathematics (Fall 2021)

*Directed Reading Program Mentor*

Spring 2023 – Spring 2025

- William Allen, *Elliptic Curve Cryptography: Data Encryption through Math* (Spring 2025)
- William Dollhopf, *Introduction to Cryptography and the ElGamal Key Cryptosystem* (Fall 2023)

- Emily Bennett, *Understanding PKC: The Diffie-Hellman Key Exchange* (Spring 2023)
- Stefan Hesseling, *RSA and Integer Factorization* (Spring 2023)

*Math Workshop Tutor*

Fall 2020 – Spring 2021

- Covers the entire selection of undergraduate math courses at Wesleyan

**Union College***Course Assistant*

Fall 2017 – Winter 2020

- MTH 340 - Linear Algebra (Winter 2020)
- MTH 110 - Calculus I (Fall 2017, Fall 2018, Fall 2019)
- MTH 112 - Calculus II (Winter 2018, Winter 2019)
- MTH 115 - Differential Vector Calculus and Matrix Theory (Spring 2018)

*Calculus Help Center Tutor*

Fall 2019 – Winter 2020

- Covers Calculus I, Calculus II, and Multivariable Calculus with Linear Algebra

**OUTREACH**

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**Project Leader (with Jeffrey Hatley)**

June 22 – July 3, 2026

*Rethinking Number Theory* [[Link](#)] [See [Paper #5](#)]

Virtual

**Project Assistant: Iwasawa Theory and Applications**

May 5-9, 2025

*Arithmetic Geometry at UNT* [[Link](#)]

Denton, TX

**Mini-Course Lecturer: Mathematical Cryptography** [[Notes](#)]

March 29-30, 2025

*Student Workshop in Mathematics* [[Link](#)]

Middletown, CT

**INVITED TALKS**

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**Joint Mathematics Meetings**

January 13, 2027

*Special Session: Explicit Methods in Number Theory*

Chicago, IL

**Bucknell University Math Colloquium**

April 3, 2026

*Bucknell University*

Lewisburg, PA

**CSUB Math Seminar**

March 27, 2026

*California State University, Bakersfield*

Bakersfield, CA

**UIC Number Theory Seminar**

March 13, 2026

*University of Illinois Chicago*

Chicago, IL

**Five College Number Theory Seminar**

February 10, 2026

*Amherst College*

Amherst, MA

**Wesleyan Algebra and Number Theory Seminar**

February 6, 2026

*Wesleyan University*

Middletown, CT

|                                                                                                                            |                                     |
|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <b>UConn Algebra Seminar</b><br><i>University of Connecticut</i>                                                           | December 3, 2025<br>Storrs, CT      |
| <b>AMS Fall Eastern Sectional Meeting</b><br><i>Special Session: Number Theory Meets Algebraic Geometry</i>                | October 25, 2025<br>Virtual         |
| <b>AMS Fall Southeastern Sectional Meeting</b><br><i>Special Session: Modular Forms in Combinatorics and Number Theory</i> | October 4, 2025<br>New Orleans, LA  |
| <b>AWM Research Symposium</b><br><i>Special Session: Curves and Abelian Varieties in Characteristic <math>p</math></i>     | May 16, 2025<br>Madison, WI         |
| <b>AMS New England Graduate Student Conference</b><br><i>Brown University</i>                                              | April 19, 2025<br>Providence, RI    |
| <b>Manhattan College Math Seminar</b><br><i>Manhattan College</i>                                                          | November 15, 2023<br>Bronx, NY      |
| <b>Union College Math Seminar</b><br><i>Union College</i>                                                                  | January 12, 2023<br>Schenectady, NY |

## CONTRIBUTED TALKS

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|---------------------------------------------------------------------------------------|----------------------------------------------|
| <b>33èmes Journées Arithmétiques</b><br><i>Université du Luxembourg</i>               | July 4, 2025<br>Belval Esch-sur-Alzette, LUX |
| <b>37<sup>th</sup> Automorphic Forms Workshop</b><br><i>University of North Texas</i> | May 4, 2025<br>Denton, TX                    |
| <b>Conférence de Théorie des Nombres Québec-Maine</b><br><i>Université Laval</i>      | October 6, 2024<br>Québec City, CAN          |
| <b>36<sup>th</sup> Automorphic Forms Workshop</b><br><i>Oklahoma State University</i> | May 24, 2024<br>Stillwater, OK               |
| <b>Graduate Student Seminar</b><br><i>Wesleyan University</i>                         | March 9, 2023<br>Middletown, CT              |
| <b>Graduate Student Seminar</b><br><i>Wesleyan University</i>                         | October 14, 2021<br>Middletown, CT           |

## SELECT WORKSHOPS AND CONFERENCES

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|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>Rethinking Number Theory<sup>+</sup></b><br><i>Distinguishing Elliptic Curves and Understanding Product Representations</i> | June 22 – July 3, 2026<br>Virtual |
| <b>Arizona Winter School</b><br><i>Computational Aspects of Arithmetic Geometry and Cryptography</i>                           | March 7-11, 2026<br>Tucson, AZ    |

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<sup>+</sup>Denotes that I served as a project leader at this workshop. See [Outreach](#) for additional details. See also [Paper #5](#).

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| <b>LMFDB, Computation, and Number Theory 2025</b><br><i>Institute for Computational and Experimental Research in Mathematics</i> | July 7-11, 2025<br>Providence, RI |
| <b>Arithmetic Geometry at UNT<sup>‡</sup></b><br><i>University of North Texas</i>                                                | May 5-9, 2025<br>Denton, TX       |
| <b>Algorithmic Number Theory Symposium XVI</b><br><i>Massachusetts Institute of Technology</i>                                   | July 15-19, 2024<br>Cambridge, MA |
| <b>Arizona Winter School</b><br><i>Abelian Varieties</i>                                                                         | March 2-6, 2024<br>Tucson, AZ     |

## SERVICE

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|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| <b>DRP Organizer (with Arianna Zikos)</b><br><i>Wesleyan Directed Reading Program</i> [ <a href="#">Link</a> ]                       | Spring 2023 – Spring 2025<br>Middletown, CT |
| <b>President</b><br><i>Wesleyan AMS Graduate Student Chapter</i>                                                                     | Fall 2023 – Spring 2025<br>Middletown, CT   |
| <b>Workshop Organizer (with Lydia Ahlstrom and Arianna Zikos)</b><br><i>Student Workshop in Mathematics</i> [ <a href="#">Link</a> ] | Spring 2025<br>Middletown, CT               |
| <b>Seminar Organizer (with Justin Bryant)</b><br><i>Wesleyan Graduate Student Seminar</i>                                            | Spring 2022 – Fall 2023<br>Middletown, CT   |
| <b>Vice President</b><br><i>Wesleyan AMS Graduate Student Chapter</i>                                                                | Fall 2021 – Spring 2023<br>Middletown, CT   |

## HONORS AND AWARDS

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|-------------------------------------------------------------------------------------------------------------|-------------------------------|
| <b>Martin Terry Resch Prize</b> [ <a href="#">Link</a> ]<br><i>Union College, Department of Mathematics</i> | May 2020<br>Schenectady, NY   |
| <b>Phi Beta Kappa Membership</b><br><i>Union College Chapter</i>                                            | May 2020<br>Schenectady, NY   |
| <b>Pi Mu Epsilon Membership</b><br><i>Union College Chapter</i>                                             | April 2019<br>Schenectady, NY |

## GRANTS

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|-------------------------------------------------------------------------------------|---------------------------------|
| <b>AMS Sectional Travel Grant</b><br><i>AMS Fall Southeastern Sectional Meeting</i> | October 2025<br>New Orleans, LA |
| <b>AWM Research Symposium Travel Grant</b><br><i>AWM Research Symposium</i>         | May 2025<br>Madison, WI         |

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<sup>‡</sup>Denotes that I served as a project assistant at this workshop. See [Outreach](#) for additional details.

## SKILLS

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**Computer:** Magma, SageMath, Python, Linux, L<sup>A</sup>T<sub>E</sub>X

**Language:** English (native), French (proficient)

## REFERENCES

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### RESEARCH

- **David Pollack**, [dpollack@wesleyan.edu](mailto:dpollack@wesleyan.edu)  
Associate Professor of Mathematics, Wesleyan University
- **Avner Ash**, [ashav@bc.edu](mailto:ashav@bc.edu)  
Professor of Mathematics, Boston College
- **Chris Williams**, [chris.williams1@nottingham.ac.uk](mailto:chris.williams1@nottingham.ac.uk)  
Assistant Professor of Pure Mathematics, University of Nottingham

### TEACHING

- **Alex Kruckman**, [akruckman@wesleyan.edu](mailto:akruckman@wesleyan.edu)  
Assistant Professor of Mathematics, Wesleyan University
- **Felipe Ramírez**, [framirez@wesleyan.edu](mailto:framirez@wesleyan.edu)  
Associate Professor of Mathematics, Wesleyan University